

Srinivas Bhootam

Data Analyst | Software Engineer

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Colony, TX, US

SUMMARY

IT professional with over 5 years of experience in AI and autonomous systems, focused on software development and data workflow optimization. Led technical teams at NVIDIA, developing Python-based tools and automating processes for computer vision and deep learning applications. Experienced in building data pipelines, creating performance dashboards using Power BI, and supporting model training and evaluation. I earned a master’s degree in business Analytics with a strong foundation in predictive modeling, data visualization, and applied data analysis. Proven ability to align business objectives with technical solutions, delivering scalable, data-driven insights that support strategic decision-making.

EDUCATION

Master of Science (M.S.), Business Analytics

Northwood University, GPA 4.0/4.0

Michigan, MI, US

08/2024 – 07/2025

Bachelor of Technology, Information Technology

Keshav Memorial Institute of Technology

Hyderabad, India

10/2013 – 05/2017

SKILLS

Programming Languages and Frameworks: Python, Node.js, JavaScript, React.js, Fast API

Web Development Technologies: HTML, CSS, XML, JSON, jQuery, Bootstrap

Database: SQL, MySQL, PostgreSQL, MongoDB, Azure SQL

AI & Machine Learning: Scikit-learn, TensorFlow, PyTorch, OpenCV, XGBoost, NLP, Time Series Forecasting, Model Deployment, Hyperparameter Tuning.

Technologies : Agile, CI/CD, Scrum, Docker, Release Management Life cycle

Tools : Jupiter Notebook, VS Code, Microsoft Azure services, AWS, GitHub, Google Collab, Posit Cloud, Power BI, Minitab, MS office, Tableau, Excel.

EXPERIENCE

Data Analyst

Hyderabad, Telangana, India

NVIDIA (via vendor)

02/2021 – 04/2024

- Led a cross-functional team delivering AI-driven solutions for NVIDIA’s autonomous vehicle systems, with a focus on deep learning, computer vision, and data pipeline optimization.
- Developed Python-based automation tools that reduced manual data tagging time by 30%, improving annotation throughput for model training.
- Designed and optimized SQL queries to extract and aggregate large datasets for model training and reporting.
- Built custom dashboards using Power BI, integrating Python logic to visualize performance metrics and support decision-making for engineering and stakeholder teams.
- Implemented workflow enhancements in annotation and validation processes, increasing throughput and improving training data quality.
- Conducted monthly performance reviews, analyzed production metrics, and maintained internal reporting systems to ensure continuous alignment with project goals.

Process Executive (Software Support, Data Operations)

Hyderabad, Telangana, India

NVIDIA (via vendor)

10/2018 – 01/2021

- Developed and maintained software tools using Python to automate data workflows and support image processing for autonomous vehicle systems.
- Built internal web-based interfaces using JavaScript and React for managing dataset labeling tasks and improving annotation efficiency.
- Implemented logic for validating and preprocessing large-scale image datasets, ensuring consistency and quality in training data.
- Contributed to the evaluation and debugging of computer vision models by identifying edge cases and assisting in classification accuracy improvements.
- Collaborated with R&D engineers to test and provide feedback on proprietary software components, contributing to performance enhancements in deep learning infrastructure.

Graduate Student – Artificial Intelligence Final Project

Midland, Michigan

Smart Voice Assistant (Astra)

- Designed and developed Astra, a smart voice assistant using **Natural Language Processing (NLP)** and **AI**.
- Integrated **OpenAI API** for real-time Q&A, and **OpenWeatherMap API** for dynamic weather updates.
- Implemented **speech recognition** and **text-to-speech** functionality to enable interactive user commands.
- Focused on creating a seamless user experience and real-time responsiveness in the assistant's design.

Graduate Student – Relational Database Management Final Project

Midland, Michigan

Smart City Utilities Management System

- Designed and implemented a Smart City Utilities Management System using **PostgreSQL** and **MongoDB Atlas**.
- Built relational schemas with key foreign relationships, views, and complex **SQL JOIN** queries for utility data analysis.
- Converted SQL schema to **NoSQL** using embedded JSON documents and wrote advanced MongoDB queries.
- Simulated a blockchain model to demonstrate secure, immutable billing records using hashing and consensus logic.
- Applied strong data modeling, schema design, and query optimization skills across all platforms.

Graduate Student – Machine Learning Capstone Project

Midland, Michigan

Smartphone Price Segmentation using ML Pipelines

- Built an end-to-end machine learning pipeline to classify smartphones, comparing five supervised models (**Random Forest, KNN, Logistic Regression, Naive Bayes, Decision Tree**) under baseline, **hyperparameter-tuned**, and **PCA-reduced** scenarios.
- Optimized Random Forest with hyperparameter tuning (89.3% accuracy) and used PCA to evaluate performance trade-offs across linear and non-linear models.
- Used **SHAP** to explain feature importance, highlighting RAM, battery, and internal memory as key pricing factors.
- Delivered interpretable, data-driven insights to support pricing strategy decisions.

Graduate Student – MSBA Capstone Project

Midland, Michigan

AI-Powered Call Center Data Automation and Forecasting analysis

- Built an interactive reporting solution for a **real-world** healthcare call center in collaboration with **Web Spark Media**.
- Conducted advanced **EDA** on call logs to identify agent performance patterns, peak hours, and appointment trends.
- **SARIMA, ARIMA, and Prophet models** applied to forecast 30-day call volumes, with SARIMA showing highest accuracy.
- Developed a **Streamlit** dashboard for live insights, including KPI tracking, agent heatmaps, and reason-for-call analysis.
- Delivered actionable recommendations to improve staffing efficiency, agent consistency, and operational planning.

Certifications

Python Data Associate - June 2025

Machine Learning, Computer Vision & AI Foundations (LinkedIn)

- Artificial Intelligence Foundations: Neural Networks
- Python for Data Science and Machine Learning Essential
- Machine Learning with Python: Foundations
- Machine Learning with Scikit-Learn
- Building Computer Vision Applications with Python

Deep Learning & Neural Networks (LinkedIn)

- Hands-On PyTorch Machine Learning
- Deep Learning Essentials
- Large Language Models: Text Classification for NLP Using BERT

Natural Language Processing (LinkedIn)

- NLP with Python for Machine Learning Essential Training

Privacy and Security - HIPAA Compliance Training – Healthcare Data Privacy and Security